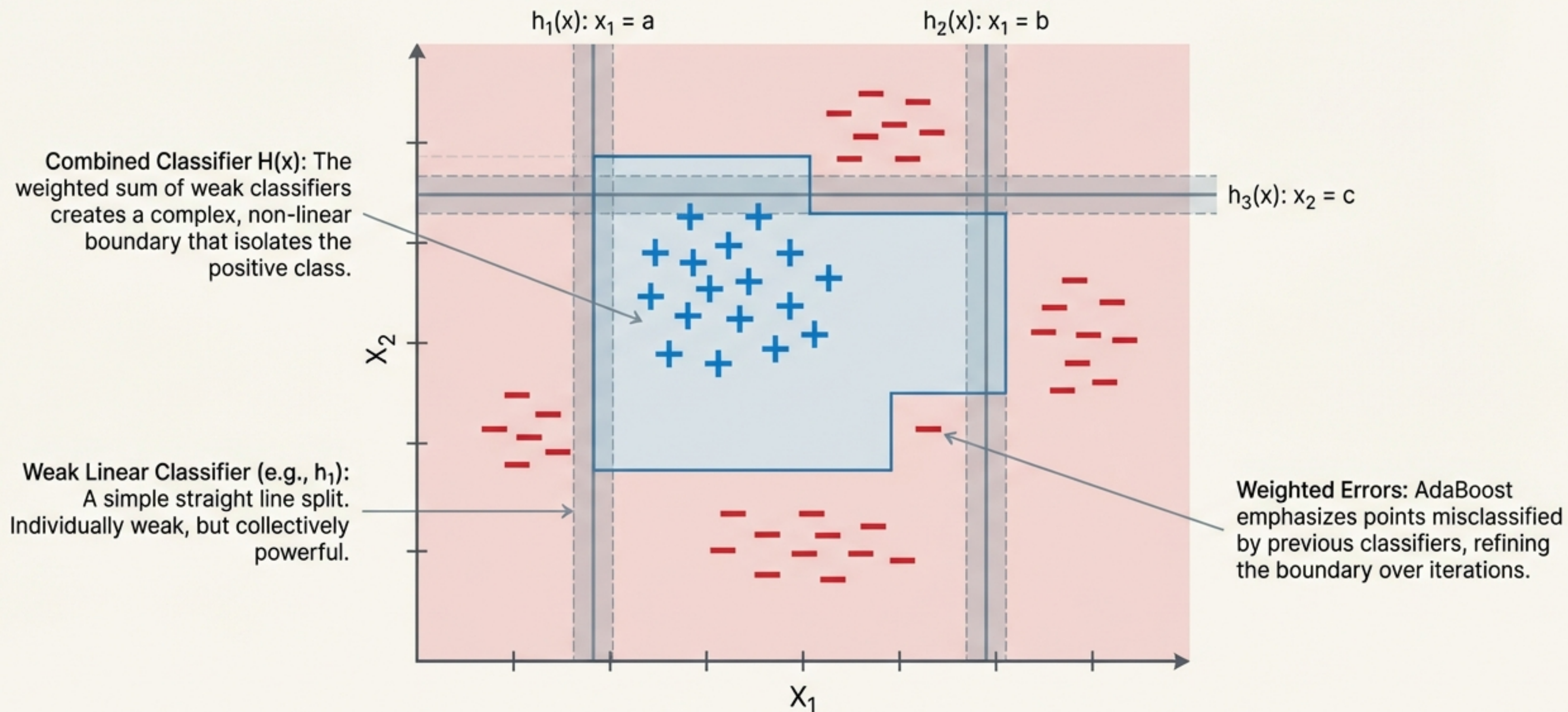
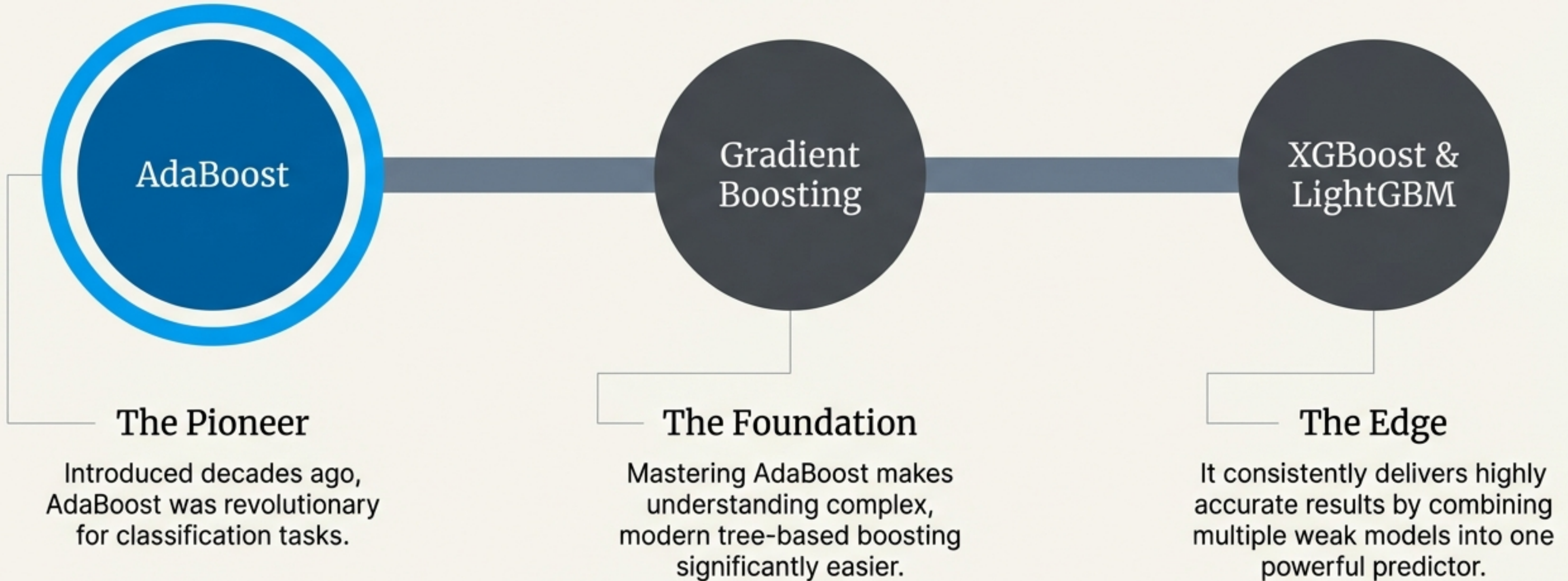


The Geometric Intuition Behind AdaBoost

Visualizing non-linear decision boundaries through the combination of simple linear classifiers.



A foundational algorithm that paved the way for modern boosting



Prerequisite 1: Understanding 'Weak Learners'



50% - Random Guessing

Definition: A machine learning model whose accuracy is barely above 50%.



51% - Just over random chance

Context: On its own, a weak learner is not very useful for robust predictions. AdaBoost's magic lies in combining hundreds of these to create a 'Strong Learner'.

Prerequisite 2: The Decision Stump is our preferred weak learner

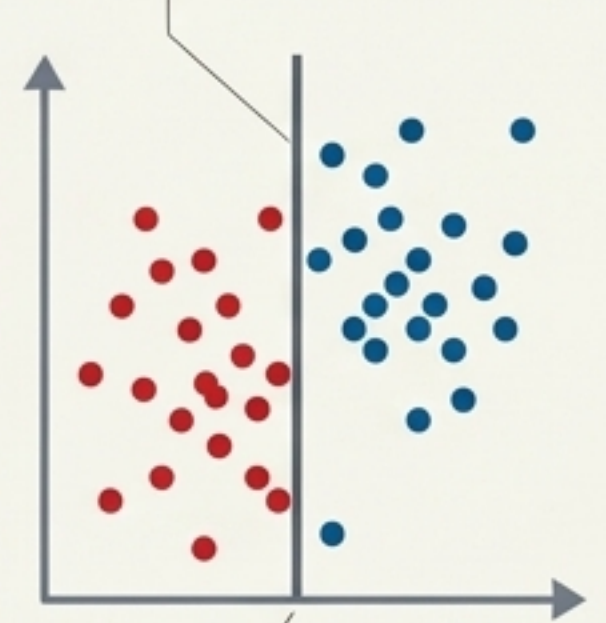
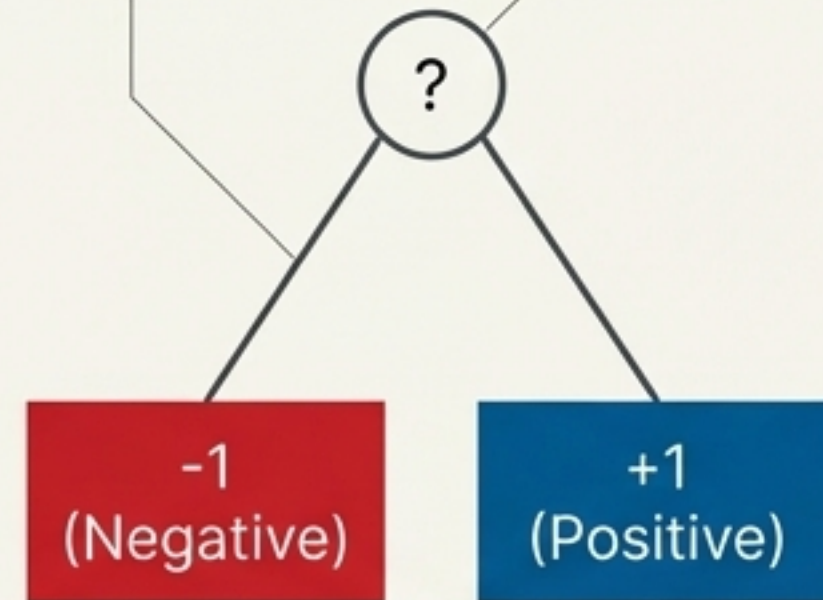
Deep Decision Tree



Decision Stump

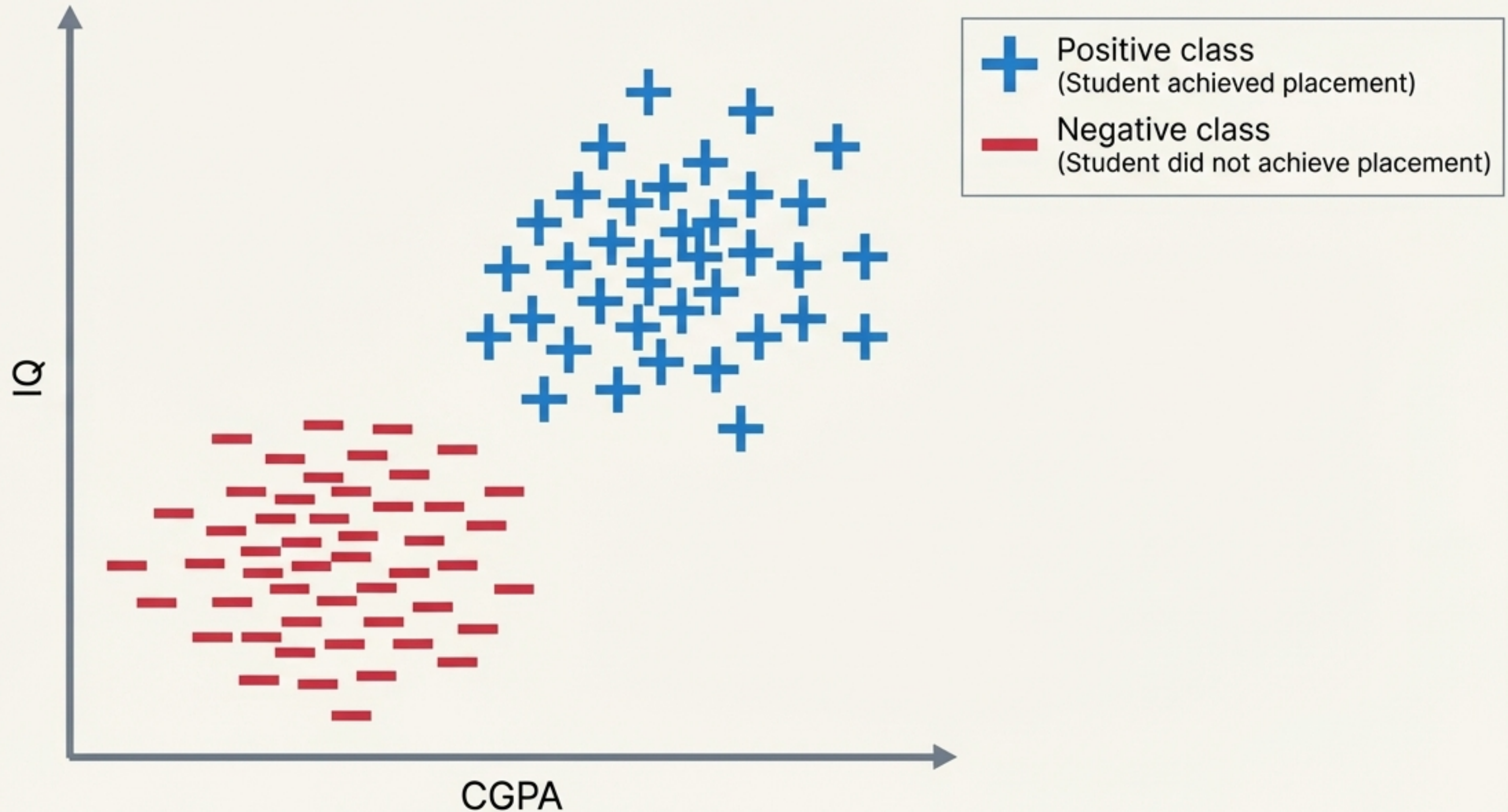
Maximum Depth of 1:
The tree can only make one split.

Single Axis: Visually represented as a single straight line on a 2D graph.



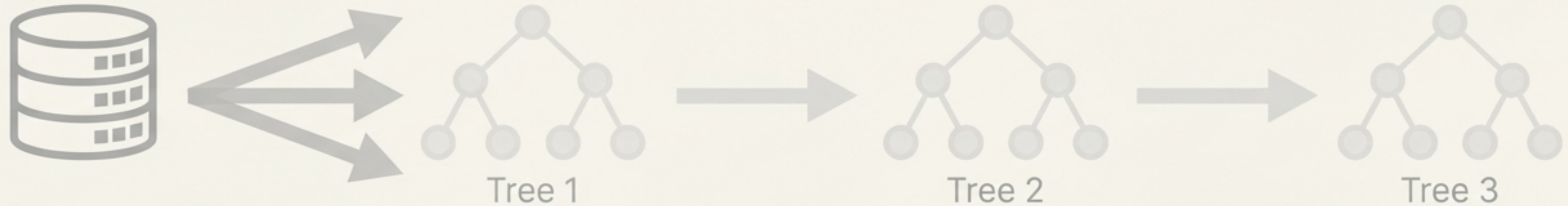
Selection Criteria:
The split is chosen based on maximum entropy reduction.

Prerequisite 3: Defining our binary targets and dataset

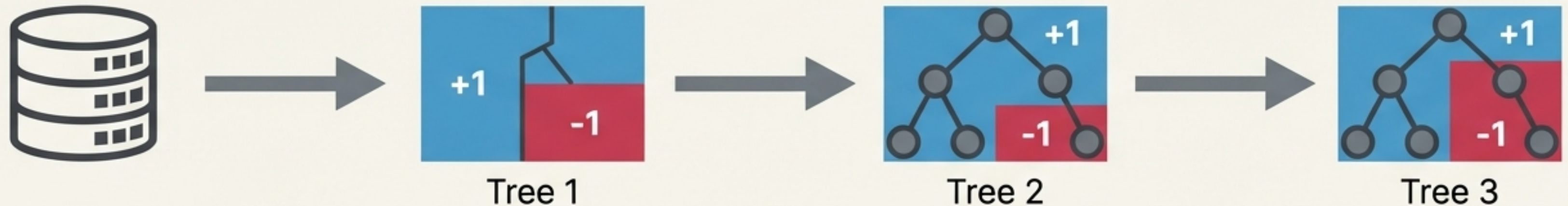


The Core Engine: Stage-wise Additive Modelling

Parallel Learning (e.g. Random Forest)

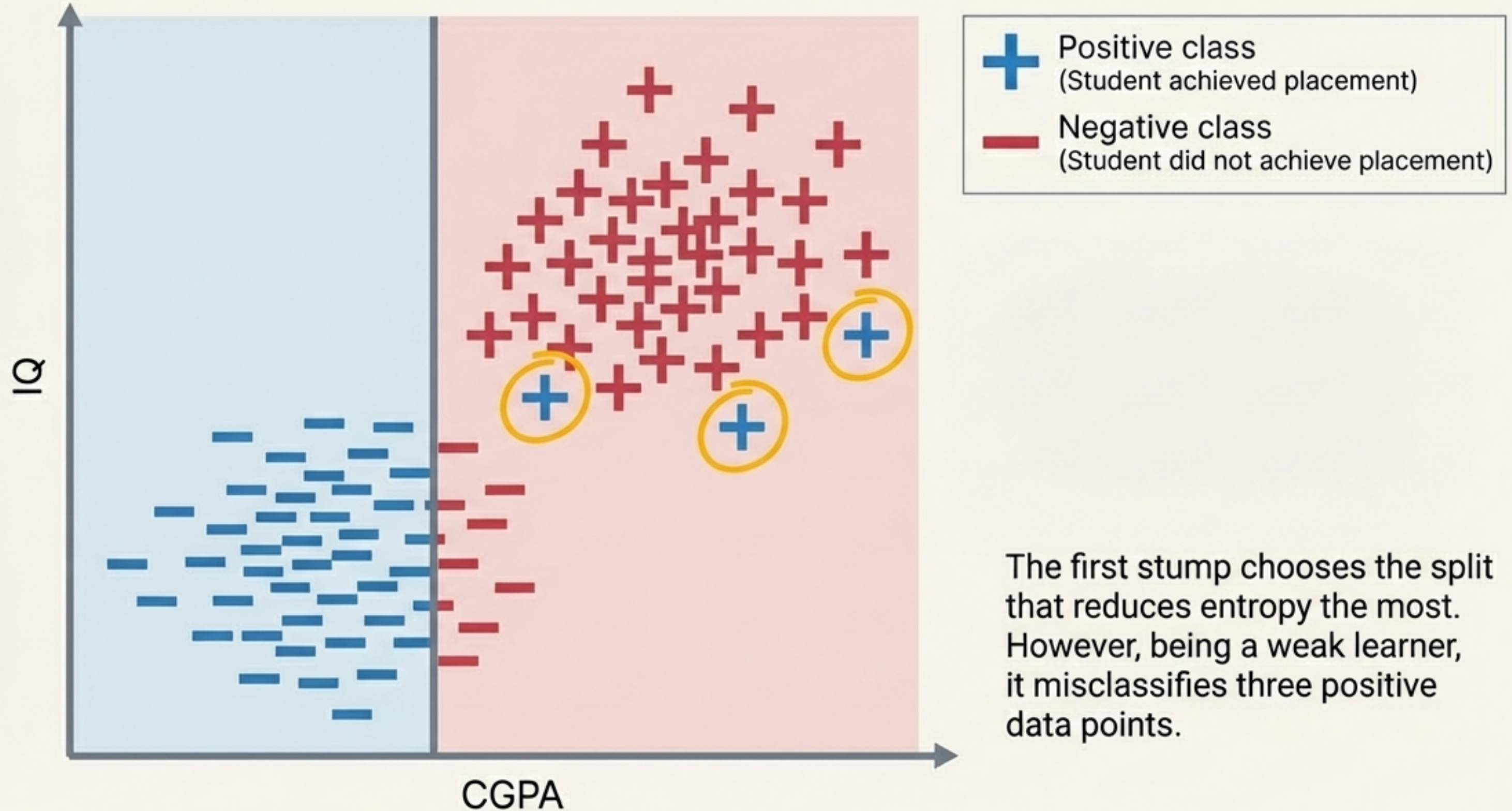


Sequential Learning (AdaBoost)

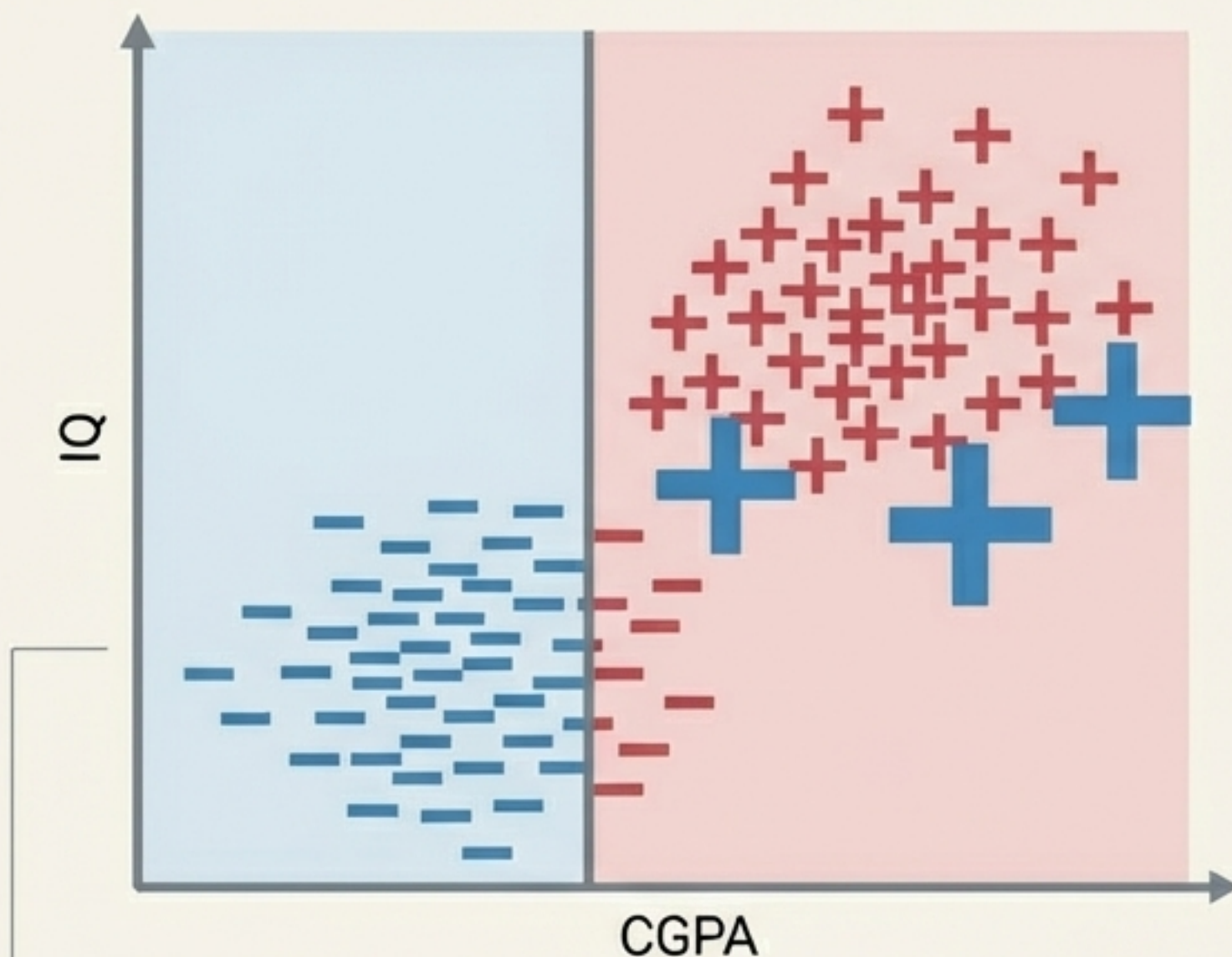


Unlike algorithms that train models in parallel (where every model has an equal vote), AdaBoost trains models sequentially. Each new model learns specifically from the mistakes of the previous one.

Stage 1: The first decision stump makes its initial cut



The Adaptation Mechanism: Reacting to mistakes



Action 1: Upsample Errors

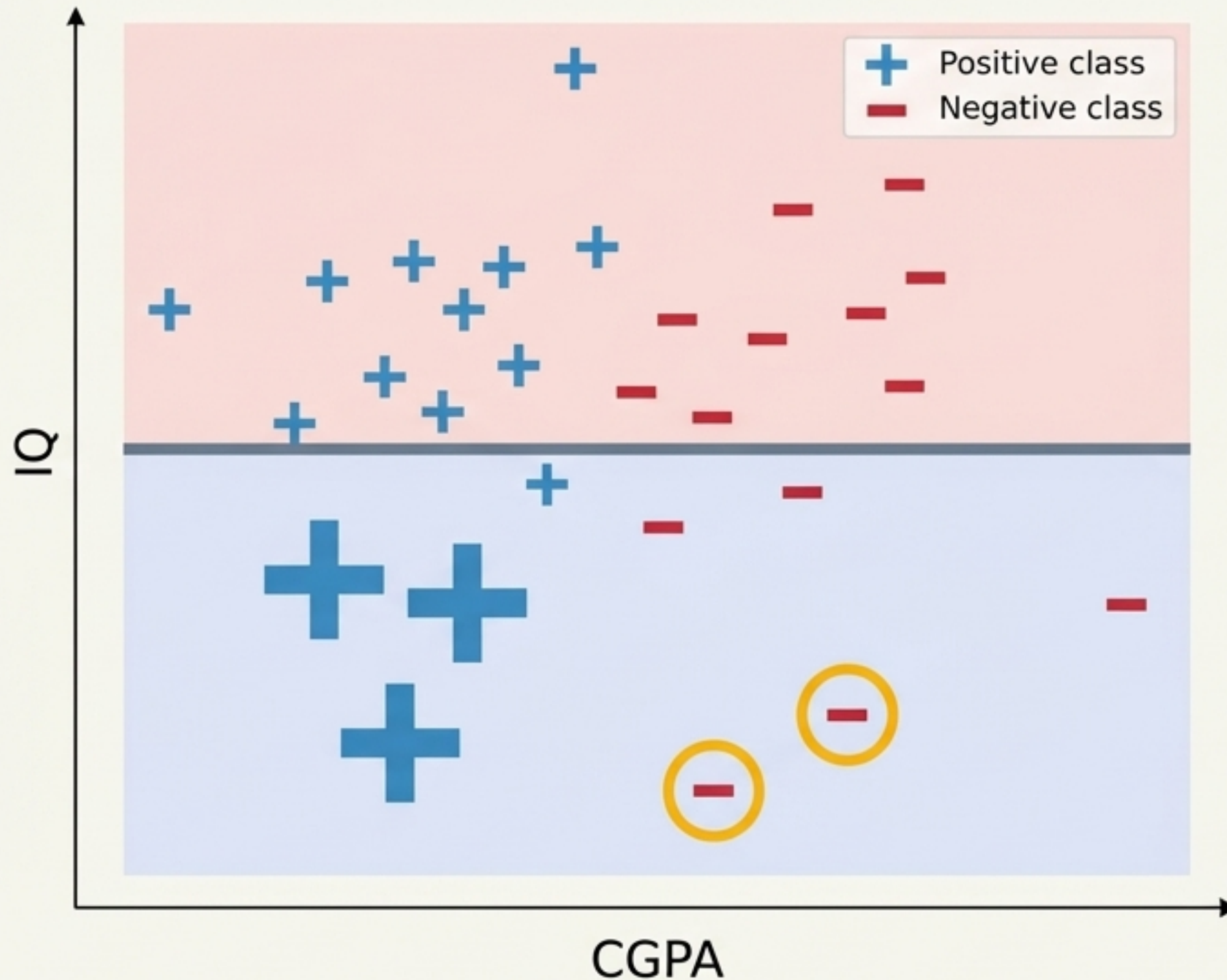
The misclassified points are given higher importance (upsampled) so the next model is forced to focus on them.



Action 2: Calculate Model Say

Unlike a standard democracy, AdaBoost weights votes based on performance. Stump 1 receives a weight based on its accuracy.

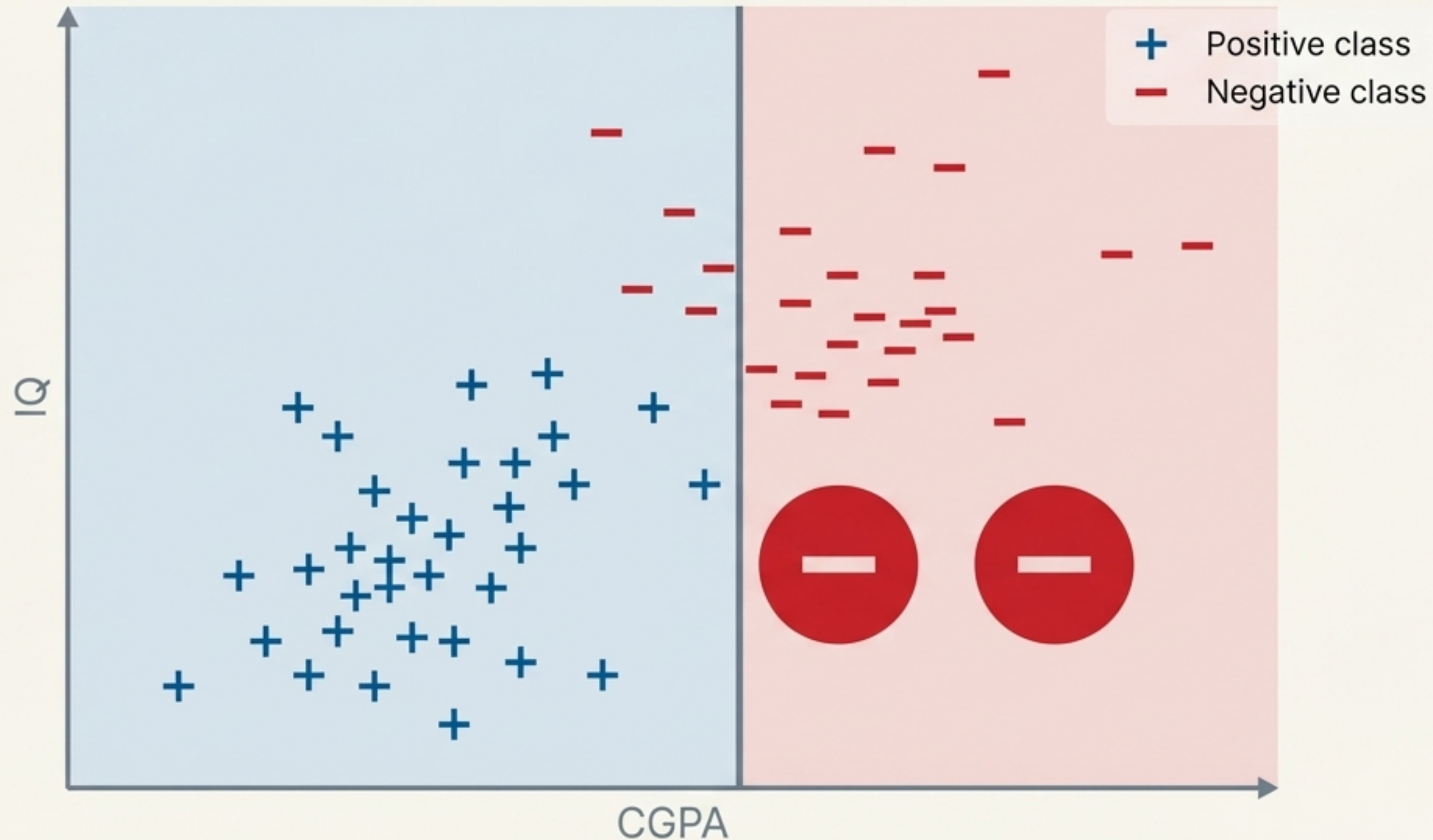
Stage 2: Shifting focus to the 'heavy' data points



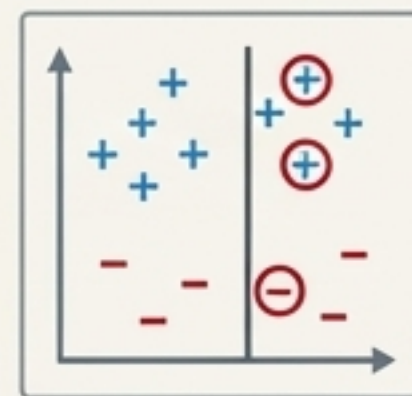
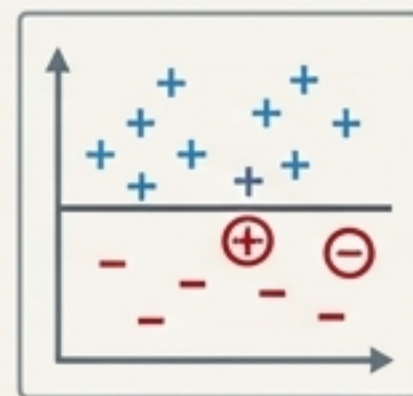
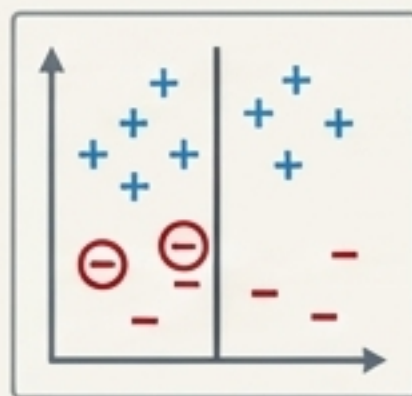
Stump 2 prioritises the upsampled points. It successfully classifies them, but misclassifies two negative points in the process. We calculate its weight (α_2) and upsample these new errors.

Stage 3: Continually refining the boundary

Stump 3 focuses on fixing the errors made by Stump 2. Its weight (α_3) is calculated based on its overall performance on this newly weighted dataset.



The Synthesis: Combining the weak learners



$$H(\mathbf{x}) = \text{sign}(\alpha_1 M_1 + \alpha_2 M_2 + \alpha_3 M_3)$$

M: The individual Decision Stump's prediction (+1 or -1).

α : The weight (say) of that specific model.

sign(): The final decision function (if positive, output +1; if negative, output -1).

Putting it into practice: Predicting a student's placement

Student Profile: CGPA 7.5, IQ 80

Model 1

$$\alpha_1 = 2$$
$$M_1 = -1$$

Model 2

$$\alpha_2 = 10$$
$$M_2 = +1$$

Model 3

$$\alpha_3 = 1$$
$$M_3 = -1$$

Calculation:

$$(2 \times -1) + (10 \times 1) + (1 \times -1)$$
$$= -2 + 10 - 1 = +7$$

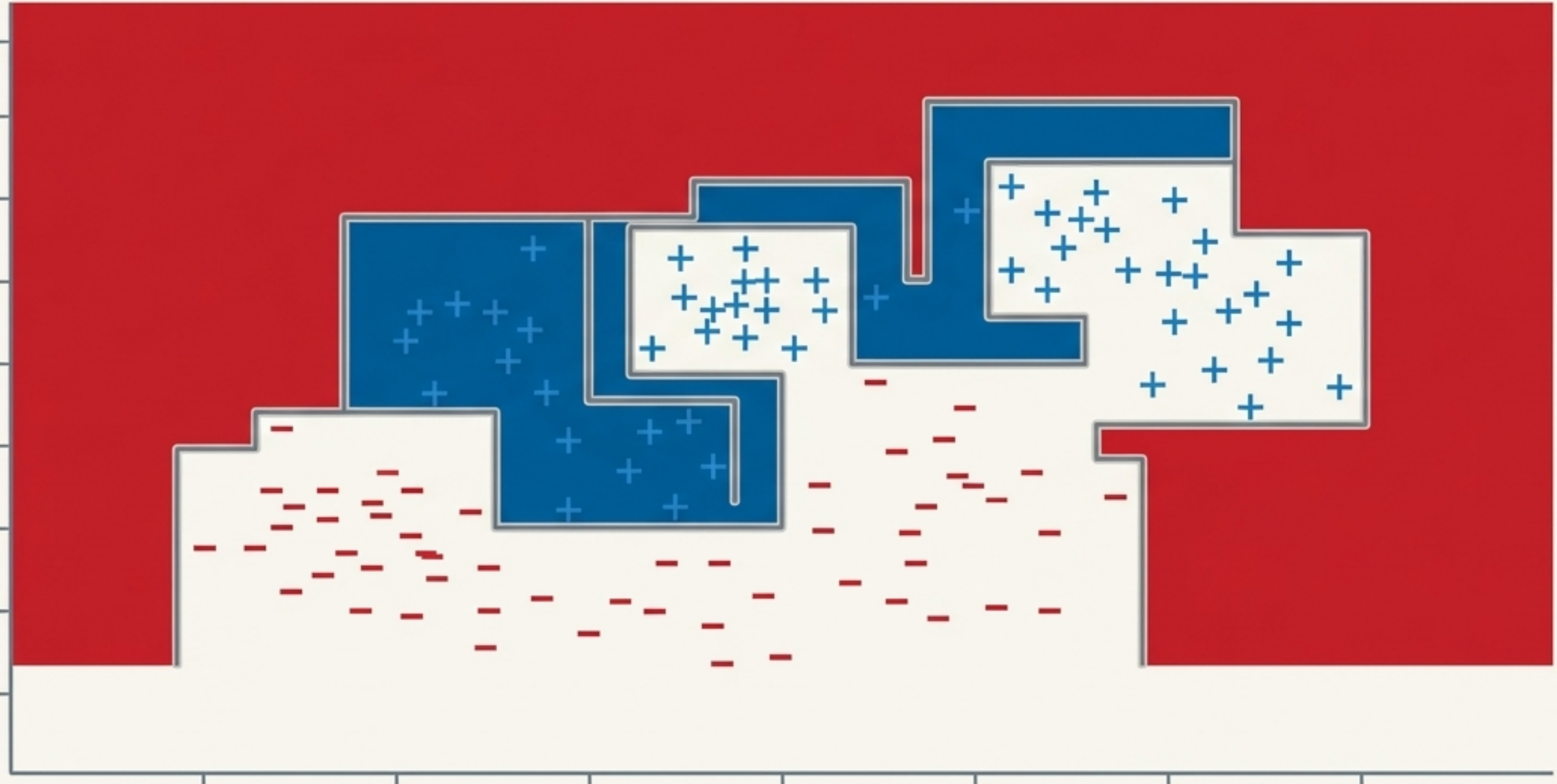
sign(+7) is Positive (+1)

+1

Result: The student is predicted to achieve placement.

The Geometric Reveal: From straight lines to complex boundaries

By overlaying multiple simple, linear decision stumps and weighting their influence, AdaBoost organically constructs a highly complex, non-linear decision boundary.



From geometric intuition to mathematical execution



Weak Learners



Upsampling Errors



Weighted Voting

Up Next:

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right)$$

We now understand the geometric intuition: AdaBoost sequentially combines weighted linear boundaries to create a non-linear classifier. Next, we must dive into the specific **formulas** driving the upsampling and α weight calculations.